

New avenues for melanoma immunotherapy: Natural Killer cells?

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Human solid malignant tumours may be particularly resistant to conventional therapies. Among solid tumours, immunological features of cutaneous melanoma have been well characterized in the past and today melanoma patients are routinely treated with the anti-immune checkpoints immunotherapy that has completely changed metastatic melanoma treatment and prognosis. Two cytotoxic cell populations may lead to the physical elimination of tumour cell targets: cytotoxic T lymphocytes (CTLs) and natural killer (NK) cells. Tumour recognition by CTLs depends on major histocompatibility complex (MHC) class I molecules, while NK cells recognize tumours expressing low or null levels of MHC class I molecules. Despite this well-established complementarity, NK cells are still left behind in the optimization of innovative immunotherapy approaches. NK cells are members of innate lymphoid cells (ILCs) that play a critical role in early host defence against invading pathogens and transformed cells. Recent findings suggest that NK cell frequencies directly correlate with the overall survival of ipilimumab-treated melanoma patients. Furthermore, in vitro and in vivo evidences indicate that NK cells can selectively kill cancer stem cells, reducing tumour size and delaying metastatic progression. The aim of this review is to provide a survey of the evidences indicating NK cells as an excellent candidate to complement the newest solid tumour immunotherapy approaches.

Immunologic approaches to acute leukemia in the elderly.

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The outcome of older patients with acute leukemia remains poor with few long-term survivors, indicating the need for treatment approaches that target pro-apoptotic pathways not influenced by chemotherapy resistance. For a long time, natural killer (NK) cells have held promise for cancer immunotherapy because, unlike T lymphocytes, they can kill tumor cells without the need for tumor-specific antigen recognition. In the treatment of acute leukemia, NK cell-based therapies have focused on in vivo expansion and activation with cytokines with only modest success. However, recent understanding of the importance of NK receptors for the recognition and lysis of leukemia cells suggests novel therapeutic strategies. The balance of inhibitory and activating signals through surface receptors, recognizing major histocompatibility complex (MHC) class I and class I-like molecules on target cells, determines whether NK cells activate killing. In this review, we discuss the biologic rationale for therapeutic strategies harnessing NK cells and focus on novel directions for their future use in elderly patients with acute leukemia.

Biology and clinical impact of human natural killer cells.

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Natural killer (NK) cells, through elaboration of cytokines and cytolytic activity, are critical to host defense against invading organisms and malignant transformation. Two subsets of human NK cells are identified according to surface CD56 expression. CD56dim cells compose the majority of NK cells and function as effectors of natural cytotoxicity and antibody-dependent cellular cytotoxicity, whereas CD56bright cells have immunomodulatory function through secretion of cytokines. For a long time, NK cells have held

promise for cancer immunotherapy because, unlike T-lymphocytes, NK cells can lyse tumor cells without tumor-specific antigen recognition. To date, NK cell therapy, largely focused on in vivo expansion and activation with cytokines, has met with only modest success. However, recent understanding of the importance of NK receptors (NKR) for recognition and lysis of tumor cells while normal cells are spared suggests novel therapeutic strategies. The balance of inhibitory and activating signals through surface receptors that recognize major histocompatibility complex class I and class I-like molecules on target cells determines whether NK cells activate killing. Identification of NKR ligands and their level of expression on normal and neoplastic cells has important implications for the rational design of immunotherapy strategies for cancer. We review recent development in the biology and clinical relevance of NK cells in cancer immunotherapy.

Recognition of viral hemagglutinins by NKp44 but not by NKp30.

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Natural killer (NK) cells destroy virus-infected and tumor cells without prior antigen stimulation. The NK cell cytotoxicity is regulated in large part by the expression of NK cell receptors that are able to bind major histocompatibility complex (MHC) class I glycoproteins. NK cells also express lysis triggering receptors specific for non-MHC ligands, including NKp30, NKp44, NKp46 and CD16. However, the nature of their ligands, recognized on target cells, is undefined. We have recently shown that the NKp46 protein, but not the CD16 protein, recognizes the hemagglutinin (HA) of influenza virus (IV) and the hemagglutinin-neuraminidase (HN) of Sendai virus (SV), and that the recognition of HA from IV requires the sialylation of NKp46 oligosaccharides. We have also demonstrated that binding of NKp46 to HA of IV is required for lysis of cells expressing the corresponding glycoproteins by a substantial subset of NK clones. Here we show that NKp44, but not NKp30, can also recognize the HA of both IV and SV and that the recognition of IV HA requires the sialylation of the NKp44 receptor in a similar way to that of NKp46. SV infection of 721.221 cells expressing MHC class I proteins resulted in the abrogation of the inhibition by NK clones expressing high levels of NKp44. In addition, the binding of NKp44 to HA improves the ability of some NK clones to lyse IV infected cells.

[Natural killer cells--biology, functions and clinical relevance].

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INTRODUCTION; Natural Killer cells (NK cells) represent the subset of peripheral lymphocytes that play critical role in the innate immune response to virus-infected and tumor transformed cells. Lysis of NK sensitive target cells could be mediated independently of antigen stimulation and without requirement of peptide presentation by the major histocompatibility complex (MHC) molecules. NK cell activity and functions are controlled by a considerable number of cell surface receptors, which exist in both inhibitory and activating isoforms. There are several groups of NK cell surface receptors: 1) killer immunoglobulin like receptors-KIR, 2) C-type lectin receptors, 3) natural cytotoxicity receptors-NCR and 4) Toll-like receptors-TLR. Defining the biology of NK cell surface receptors has contributed to the concept of the manner how NK cells selectively recognize and lyse tumor and virally infected cells while sparing normal cells. Further, identification of NK receptor ligands and their expression on the normal and transformed cells has led to the development of clinical approaches to manipulating receptor/ligand interactions that showed clinical benefit. NK cells are the first lymphocyte subset that reconstitute the peripheral blood

following allogeneic HSCT and multiple roles for alloreactive donor NK cells have been demonstrated, in diminishing Graft vs. Host Disease (GvHD) through selective killing recipient dendritic cells, prevention of graft rejection by killing recipient T cells and participation in Graft vs. Leukaemia (GvL) effect through destruction of residual host tumor cells. Besides their role in HSCT, NK cell receptors have an important clinical relevance that reflects from the fact that they play a crucial role in the development of some diseases as well as in possibilities of managing all NK receptors through selective expansion and usage of NK cells in cancer immunotherapy.

Efficient generation of gene-modified human natural killer cells via alpharetroviral vectors.

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Natural killer (NK) cells play an important role in tumor immunotherapy with their unique capability of killing transformed cells without the need for prior sensitization and without major histocompatibility complex (MHC)/peptide restriction. However, tumor cells can escape NK cell cytotoxicity by various tumor immune escape mechanisms. To overcome these escape mechanisms, NK cells can be modified to express chimeric antigen receptors (CARs), enhancing their tumor-specific cytotoxicity. To determine the most efficacious method to modify human NK cells, we compared different retroviral vector systems, retroviral pseudotypes, and transduction protocols. Using optimized transduction conditions, the highest transduction levels (up to 60%) were achieved with alpharetroviral vectors. Alpharetroviral-modified primary human NK cells exhibited no alteration in receptor expression and had similar degranulation activity as untransduced NK cells, thus demonstrating that alpharetroviral modification did not negatively affect NK cell cytotoxicity. Transduction of NK cells with an alpharetroviral vector containing a CD19 CAR expression cassette selectively enhanced NK cell cytotoxicity towards CD19-expressing leukemia cells, achieving nearly complete elimination of leukemia cells after 48 h. Taken together, alpharetroviral vectors are promising tools for NK cell-mediated cancer immunotherapy applications. Efficient modification of human NK cells using alpharetroviral vectors. Anti-CD19-CAR-NK cells exhibited improved cytotoxicity towards CD19(+) leukemia cells. Alpharetroviral vectors are promising tools for immunotherapy applications using NK cells.

Natural killer cells: the journey from puzzles in biology to treatment of cancer.

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Natural Killer (NK) cells are innate immune effectors that are primarily involved in immunosurveillance to spontaneously eliminate malignantly transformed and virally infected cells without prior sensitization. NK cells trigger targeted attack through release of cytotoxic granules, and secrete various cytokines and chemokines to promote subsequent adaptive immune responses. NK cells selectively attack target cells with diminished major histocompatibility complex (MHC) class I expression. This "Missing-self" recognition by NK cells at first puzzled researchers in the early 1990s, and the mystery was solved with the discovery of germ line encoded killer immunoglobulin receptors that recognize MHC-I molecules. This review summarizes the biology of NK cells detailing the phenotypes, receptors and functions; interactions of NK cells with dendritic cells (DCs), macrophages and T cells. Further we discuss the various strategies to modulate NK cell activity and the practice of NK cells in cancer immunotherapy employing NK cell lines,

autologous, allogeneic and genetically engineered cell populations.

Natural killer cells and cancer.

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Natural killer (NK) cells are lymphocytes that were first identified for their ability to kill tumor cells without deliberate immunization or activation. Subsequently, they were also found to be able to kill cells that are infected with certain viruses and to attack preferentially cells that lack expression of major histocompatibility complex (MHC) class I antigens. The recent discovery of novel NK receptors and their ligands has uncovered the molecular mechanisms that regulate NK activation and function. Several activating NK cell receptors and costimulatory molecules have been identified that permit these cells to recognize tumors and virus-infected cells. These are modulated by inhibitory receptors that sense the levels of MHC class I on prospective target cells to prevent unwanted destruction of healthy tissues. In vitro and in vivo, their cytotoxic ability can be enhanced by cytokines, such as interleukin (IL)-2, IL-12, IL-15 and interferon alpha/beta (IFN-alpha/beta). In animal studies, they have been shown to play a critical role in the control of tumor growth and metastasis and to provide innate immunity against infection with certain viruses. Following activation, NK cells release cytokines and chemokines that induce inflammatory responses; modulate monocyte, dendritic cells, and granulocyte growth and differentiation; and influence subsequent adaptive immune responses. The underlining mechanism of discriminating tumor cells and normal cells by NK cells has provided new insights into tumor immunosurveillance and has suggested new strategies for the treatment of human cancer.

Natural killer cells in human cancer: from biological functions to clinical applications.

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Natural killer (NK) cells are central components of the innate immunity. In murine models, it has been shown that NK cells can control both local tumor growth and metastasis due to their ability to exert direct cellular cytotoxicity without prior sensitization and to secrete immunostimulatory cytokines like IFN- γ . The latter participates in cancer elimination by inhibiting cellular proliferation and angiogenesis, promoting apoptosis, and stimulating the adaptive immune system, and it is instrumental for enhancing Ag processing and presentation. Nevertheless, NK cells display impaired functionality and capability to infiltrate tumors in cancer patients. Also, NK cells are feasible targets of stimulation to participate in immunotherapeutic approaches like antibody-based strategies and adoptive cell transfer. Thus, multiple attempts currently aim to manipulate NK for utilization in the immunotherapy of cancer.

Overcoming Resistance to Checkpoint Inhibitors: Natural Killer Cells in Non-Small Cell Lung Cancer.

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Immune checkpoint inhibitors (ICIs) have revolutionized cancer treatments over the last 10 years, with even increasing indications in many neoplasms. Non-small cell lung cancer (NSCLC) is considered highly

immunogenic, and ICIs have found a wide set of applications in this area, in both early and advanced lines of treatment, significantly changing the prognosis of these patients. Unfortunately, not all patients can benefit from the treatment, and resistance to ICIs can develop at any time. In addition to T lymphocytes, which are the major target, a variety of other cells present in the tumor microenvironment (TME) act in a complex cross-talk between tumor, stromal, and immune cells. An imbalance between activating and inhibitory signals can shift TME from an "anti-" to a "pro-tumorigenic" phenotype and vice versa. Natural killer cells (NKs) are able to recognize cancer cells, based on MHC I (self and non-self) and independently from antigen presentation. They represent an important link between innate and adaptive immune responses. Little data are available about the role of pro-inflammatory NKs in NSCLC and how they can influence the response to ICIs. NKs express several ligands of the checkpoint family, such as PD-1, TIGIT, TIM-3, LAG3, CD96, IL1R8, and NKG2A. We and others have shown that TME can also shape NKs, converting them into a pro-tumoral, pro-angiogenic "nurturing" phenotype through "decidualization." The features of these NKs include expression of CD56, CD9, CD49a, and CXCR3; low CD16; and poor cytotoxicity. During ICI therapy, tumor-infiltrating or associated NKs can respond to the inhibitors or counteract the effect by acting as pro-inflammatory. There is a growing interest in NKs as a promising therapeutic target, as a basis for adoptive therapy and chimeric antigen receptor (CAR)-NK technology. In this review, we analyzed current evidence on NK function in NSCLC, focusing on their possible influence in response to ICI treatment and resistance development, addressing their prognostic and predictive roles and the rationale for exploiting NKs as a tool to overcome resistance in NSCLC, and envisaging a way to repolarize decidual NK (dNK)-like cells in lung cancer.
